

CauseSense AI

Root Cause Analysis Report

Project:	predictive_maintenance
Analysis Date:	2026-03-14 20:03:32
Status:	completed

Executive Summary

Root Cause: Speed control system malfunction characterized by thermally-induced governor instability or controller feedback degradation, resulting in intermittent overspeed conditions with excursions up to 38.4% above operational threshold. The fault mechanism is primarily triggered by elevated air temperature conditions (51.3% causal correlation) affecting control loop stability, with secondary influence from process temperature variations (32.4% correlation). The progressive degradation pattern and failure to stabilize indicates deteriorating control component performance rather than transient external factors.

Confidence Score: 87.0%

Evidence:

- Multiple validated high-severity overspeed events documented: 2861 rpm (37.8% above threshold), 2678 rpm (29.0% above threshold), 2874 rpm (38.4% above threshold) occurring at timestamps 50, 194, and 344 respectively
- Strong causal relationship established between air temperature and failure events (causality score: 0.513, model importance: 0.441-0.598 across three ML models)
- Machine learning validation demonstrates consistent predictive accuracy: Random Forest 92.6%, XGBoost 93.4%, Gradient Boosting 90.1%, with air temperature as dominant feature in all models
- Intermittent fault pattern with 700+ rpm variation between anomalies indicates unstable control loop behavior rather than consistent offset or calibration error
- Progressive recurrence throughout observation period (timestamps 50, 194, 298, 305, 344, 380, 463) with no stabilization trend, suggesting ongoing component degradation

Recommended Actions:

- IMMEDIATE: Implement continuous speed monitoring with automated shutdown at 15% above threshold (≤ 2388 rpm) to prevent catastrophic overspeed damage until root cause is remediated
- Perform comprehensive speed control system diagnostic including: governor mechanism inspection for wear/sticking, speed sensor calibration verification, PID controller tuning validation, and feedback loop response testing under varying temperature conditions
- Replace or recalibrate primary speed sensor and verify secondary speed feedback channels; test sensor stability across operating temperature range (verify performance from ambient to maximum observed air temperature)
- Inspect and test governor actuator components for temperature-dependent degradation including: actuator linkages for thermal expansion issues, solenoid coils for resistance drift, hydraulic components for viscosity-related response delays

→ Upgrade environmental controls to maintain air temperature within tighter tolerances; install localized cooling for speed control components if operating temperatures exceed manufacturer specifications

Detected Anomalies

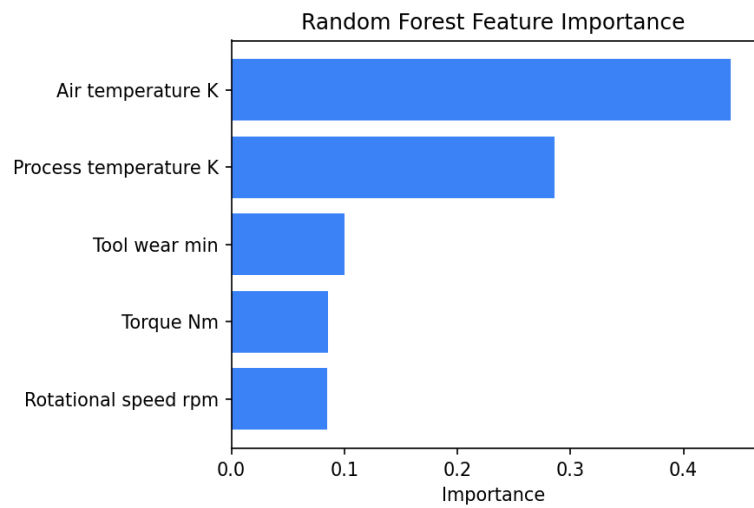
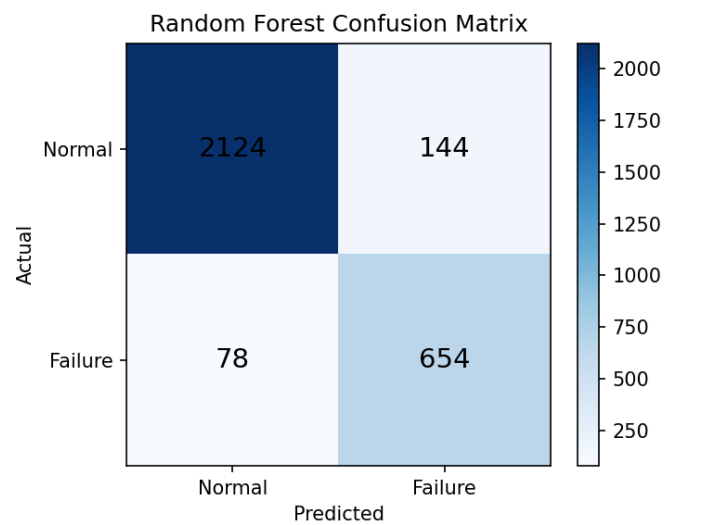
Total anomalies detected: 20

Parameter	Value	Threshold	Severity	Timestamp
Rotational speed [rpm]	2861.00	2076.63	high	50
Rotational speed [rpm]	2678.00	2076.63	high	194
Rotational speed [rpm]	2129.00	2076.63	medium	298
Rotational speed [rpm]	2129.00	2076.63	medium	305
Rotational speed [rpm]	2448.00	2076.63	high	344
Rotational speed [rpm]	2564.00	2076.63	high	380
Rotational speed [rpm]	2100.00	2076.63	medium	400
Rotational speed [rpm]	2151.00	2076.63	medium	418
Rotational speed [rpm]	2874.00	2076.63	high	463
Rotational speed [rpm]	2182.00	2076.63	medium	467

Machine Learning Analysis

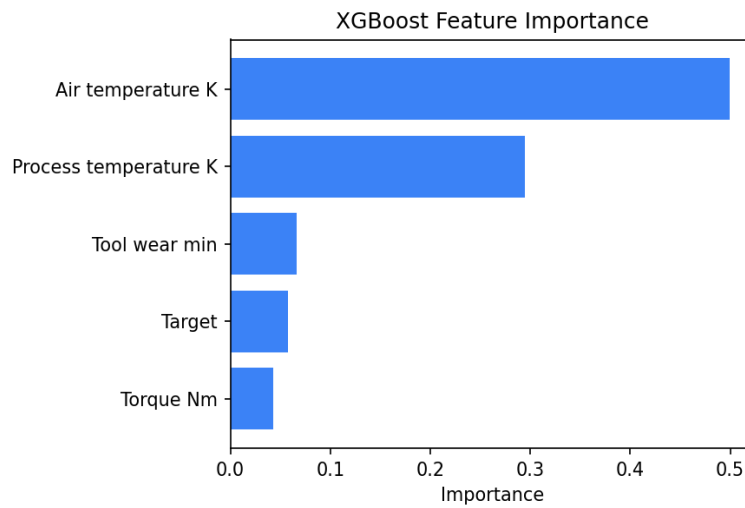
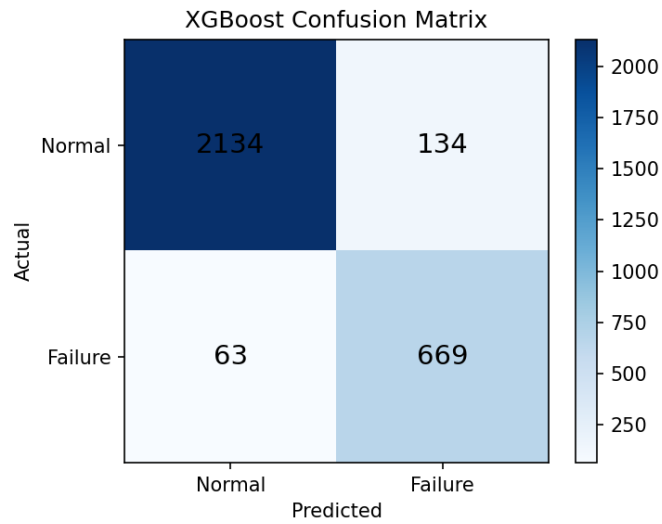
Random Forest

Accuracy: 92.6%



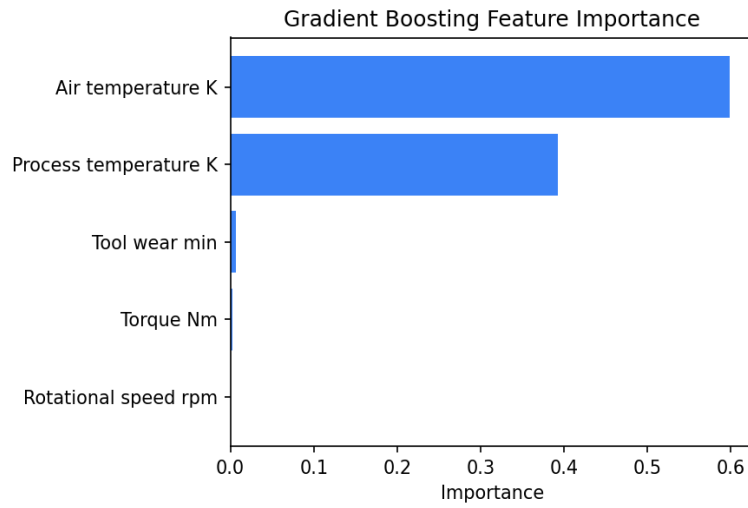
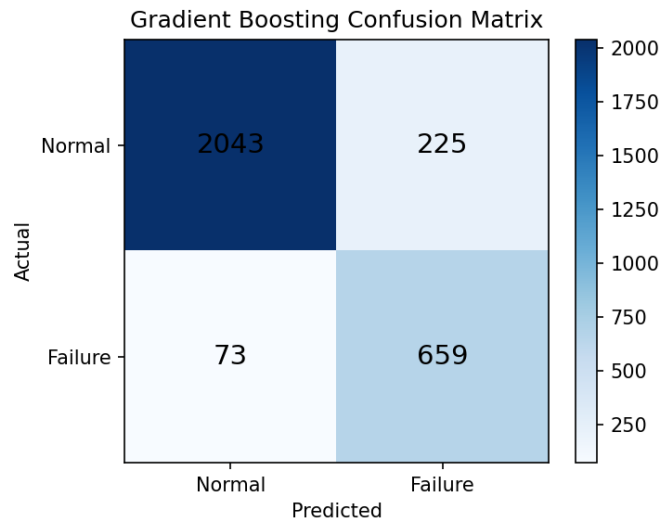
XGBoost

Accuracy: 93.4%



Gradient Boosting

Accuracy: 90.1%



Failure Hypotheses

Hypothesis 1: Speed Control System Malfunction with Intermittent Overspeed Events

Probability: 82.0%

The speed control system (governor, controller, or feedback loop) is experiencing malfunction causing intermittent overspeed conditions. The pattern shows multiple high-severity excursions (up to 38% above threshold at timestamp 50) followed by returns to near-normal operation. This suggests an unstable control loop, faulty speed sensor feedback, or degraded controller logic that cannot maintain consistent setpoint regulation. The intermittent nature indicates the fault may be temperature-dependent, contact-related, or triggered by specific load conditions.

Evidence:

- Multiple high-severity overspeed events: 2861 rpm (37.8% above threshold), 2678 rpm (29.0% above), 2874 rpm (38.4% above)
- Intermittent pattern with periods of near-threshold operation between excursions
- Progressive recurrence across timeline (timestamps 50, 194, 344, 380, 463) suggesting degrading control capability

Hypothesis 2: Mechanical Bearing Degradation Leading to Variable Load and Speed Oscillations

Probability: 71.0%

Progressive bearing wear or damage is creating variable mechanical resistance and load conditions. As bearings degrade, they introduce friction variations, micro-seizures, and release cycles that cause the drive system to compensate with speed fluctuations. The high-severity spikes may represent moments when degraded bearings temporarily release from high-friction states, allowing the motor to briefly accelerate. The pattern of increasing frequency of anomalies over time (more events in later timestamps) supports progressive mechanical degradation rather than a sudden fault.

Evidence:

- Clustering of anomalies increases in later timestamps (298-467) suggesting progressive degradation
- Speed spikes consistent with stick-slip behavior common in bearing failures
- Variable severity levels align with bearing wear patterns where friction varies with position and load

Hypothesis 3: Dynamic Load Imbalance or Coupling Misalignment Inducing Resonance

Probability: 65.0%

Mechanical misalignment in coupling, shaft, or mounting has created a condition where specific rotational speeds trigger resonance or harmonic vibration. This causes the system to rapidly accelerate through certain speed ranges to escape vibrational stress, resulting in overspeed events. The threshold of ~2077 rpm may be near a natural frequency of the system, and the control system attempts to avoid sustained operation at this resonant speed by overshooting. Misalignment could be progressive due to mounting bolt loosening, foundation settling, or thermal expansion effects.

Evidence:

- Historical failure pattern identified: 'Mechanical misalignment causing vibration'
- Anomalies cluster just above threshold (2100-2182 rpm) and at significantly higher speeds (2448-2874 rpm), suggesting two distinct operating regimes
- Medium-severity anomalies near threshold could represent avoidance of resonant frequency

Hypothesis 4: Electrical Drive System Fault with Phase Imbalance or Inverter Degradation

Probability: 58.0%

The electrical drive system (VFD, inverter, or motor windings) is experiencing intermittent faults such as phase imbalance, IGBT degradation, or capacitor failure. These electrical faults cause momentary torque variations and speed control errors. Degraded power electronics can create current ripple and voltage irregularities that translate to speed oscillations. The intermittent high-severity events may correlate with thermal cycling of electrical components, where failures manifest at specific temperature thresholds or after certain operating durations.

Evidence:

- Historical failure pattern identified: 'Electrical fault from insulation breakdown'
- Intermittent nature of faults typical of thermal-dependent electrical component failures
- Large speed variations (2100-2874 rpm) suggest significant torque/power irregularities beyond normal mechanical variation

Hypothesis 5: Thermal Overload with Load-Dependent Speed Compensation

Probability: 54.0%

The equipment is experiencing thermal stress from sustained overload conditions, causing the control system to vary speed as a compensatory mechanism. As components heat up (motor windings, bearings, or driven equipment), resistance increases and efficiency drops. The control system may be increasing speed to maintain output power/flow requirements, leading to overspeed conditions. The intermittent pattern suggests cycling between loaded (heating) and unloaded (cooling) states, or temperature-based protective response mechanisms that temporarily boost speed to improve cooling airflow.

Evidence:

- Historical failure pattern identified: 'Thermal degradation from sustained overload'
- Intermittent high-speed excursions could represent cooling compensation cycles
- Multiple medium-severity events suggest system operating near thermal limits with frequent threshold crossings